

Experiment No. 13

Switching of Electrical Loads Using Relays

Objective:

To understand the switching to Electrical load using Relays

Apparatus:

- A Jumper Wires
- Switching of Electrical Load Trainer

Introduction:

The advantage of relays is that it takes a relatively small amount of power to operate the relay coil. However, a relay switch circuit can be used to control motors, heaters, lamps or AC circuits which themselves can draw a lot more electrical voltage, current and therefore power.

The electro-mechanical relay is an output device (actuator) which come in a whole host of shapes, sizes and designs, and have many uses and applications in electronic circuits. But while electrical relays can be used to allow low power electronic or computer type circuits to switch relatively high currents or voltages both “ON” or “OFF”, some form of relay switch circuit is required to control it.

Electromechanical Relays:

Relays are electromechanical devices that use an electromagnet to operate a pair of movable contacts from an open position to a closed position. The relay is a switch that is controlled by a small electric current. It can be used to control motors, heaters or lamps

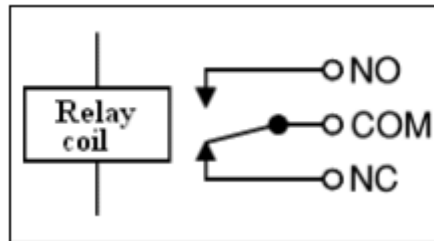


Figure 1 Circuit Symbol for a relay

circuits which themselves can draw a lot more electrical power. Figure 1 shows the circuit symbol for a relay.

In figure 2, the relay's coil is energized by the low-voltage (12 VDC) source, while the singlepole, single-throw (SPST) contact interrupts the high-voltage (480 VAC) circuit. It is quite likely that the current required to energize the relay coil will be hundreds of times less than the current rating of the contact. Typical relay coil currents are well below 1 amp, while typical contact ratings for industrial relays are at least 10 amps.

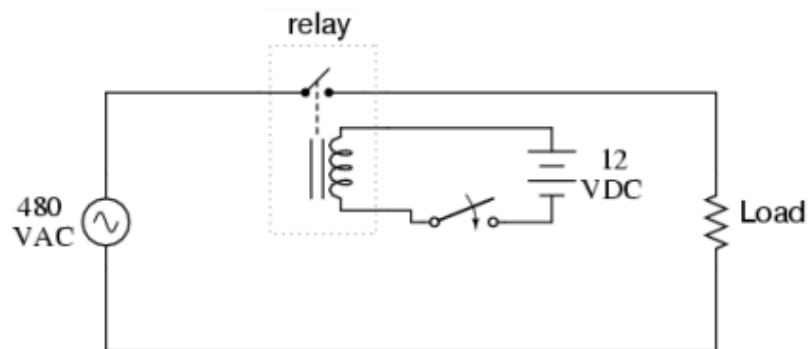


Figure 2 Relay drive an AC load

One relay coil/armature assembly may be used to actuate more than one set of contacts. Those contacts may be normally-open, normally-closed, or any combination of the two. As with switches, the “normal” state of a relay's contacts is that state when the coil is de-energized and not connected to any circuit. Figure 3 shows different relay's contacts arrangement.

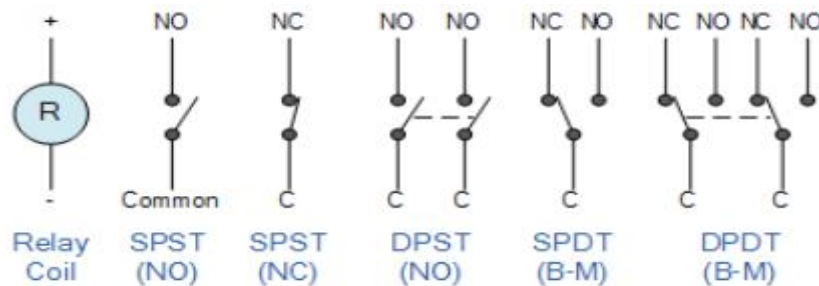


Figure 3 Different Relay's Contacts Arrangement

Relay contacts may be open-air pads of metal alloy, mercury tubes, or even magnetic reeds. Consider these factors when choosing a relay for use in industrial controls:

1. Voltages driving loads are the first concern. The voltage rating of a relay must be greater than or equal to the voltage driving the load. The frequency of the switched voltage is also critical. Because ac current fluctuates from positive to negative crossing through zero, the switched voltage will vary between the maximum voltage and zero. Dc voltage, on the other hand, is always at the maximum value, causing maximum wear on the contacts with every switch.
2. The current required depends on the type of load. Most loads don't draw a constant current. In fact, the current demand of most loads varies somewhat predictably.

Transistors:

The design and types of relays switching circuits is huge, but many small electronic projects use transistors as their main switching device as the transistor can provide fast switching (ON-OFF) control. There are different types of transistors:

The Bipolar Junction Transistor (BJT): is a three-layer device constructed from two semiconductor diode junctions joined together, one forward biased and one reverse biased. There are two main types of bipolar junction transistors (BJT) the NPN and the PNP transistor, see figure 4. These transistors are “Current Operated Devices” where a much smaller base current causes a larger emitter to collector current, which themselves are nearly equal, to flow.

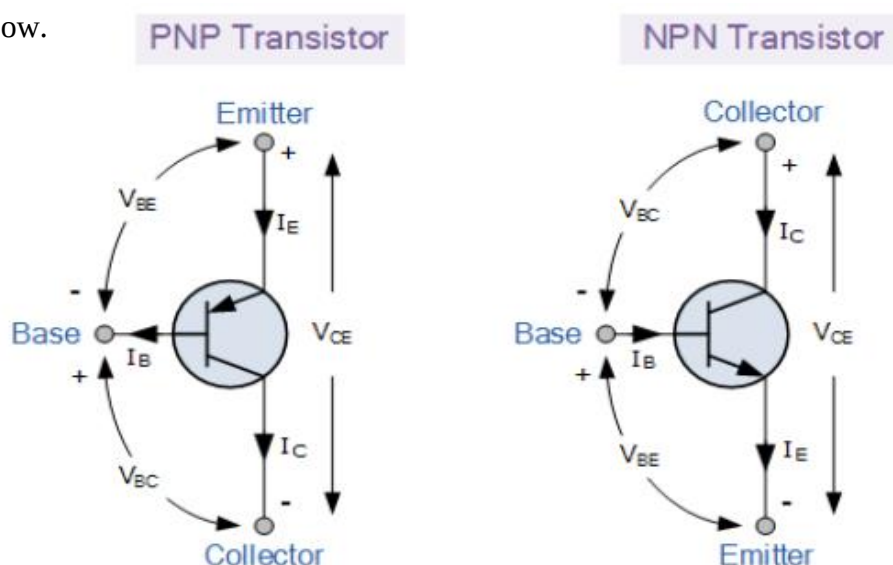
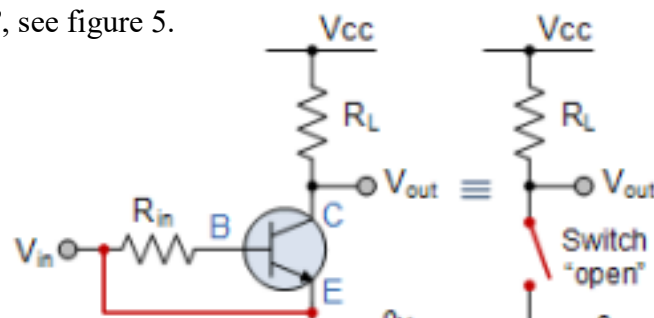


Figure 4 The two types of BJT

Bipolar Junction Transistors (BJT's) can also be used as an electronic switch between their saturation and cut-off regions to control devices such as lamps, motors and solenoids etc. If Inductive loads such as DC motors, relays and solenoids are used, they require a “freewheeling” diode placed across the load. This helps prevent any induced back emf's generated when the load is switched “OFF” from damaging the transistor. Let us consider npn bipolar junction transistor for understanding the cut-off and saturation regions:

Cut-Off Region:

Here the operating conditions of the transistor are zero input base current (I_B), zero output collector current (I_C) and maximum collector voltage (V_{CE}) which results in a large depletion layer and no current flowing through the device. Therefore, the transistor is switched “Fully-OFF”, see figure 5.



Then we can define the “cut-off region” or “OFF mode” when using a bipolar transistor as a switch as being, both junctions reverse biased, $V_B < 0.7v$ and $I_C = 0$.

Saturation Region:

Here the transistor will be biased so that the maximum amount of base current is applied, resulting in maximum collector current resulting in the minimum collector emitter voltage drop which results in the depletion layer being as small as possible and maximum current flowing through the transistor. Therefore, the transistor is switched “Fully-ON”, see figure 6.

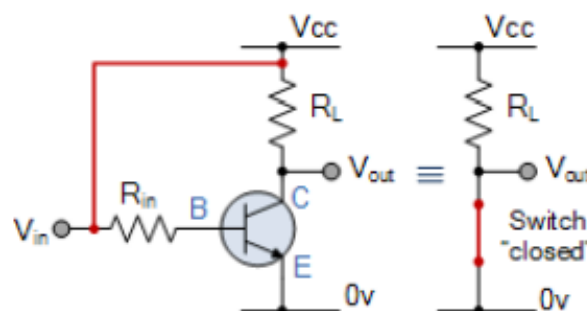
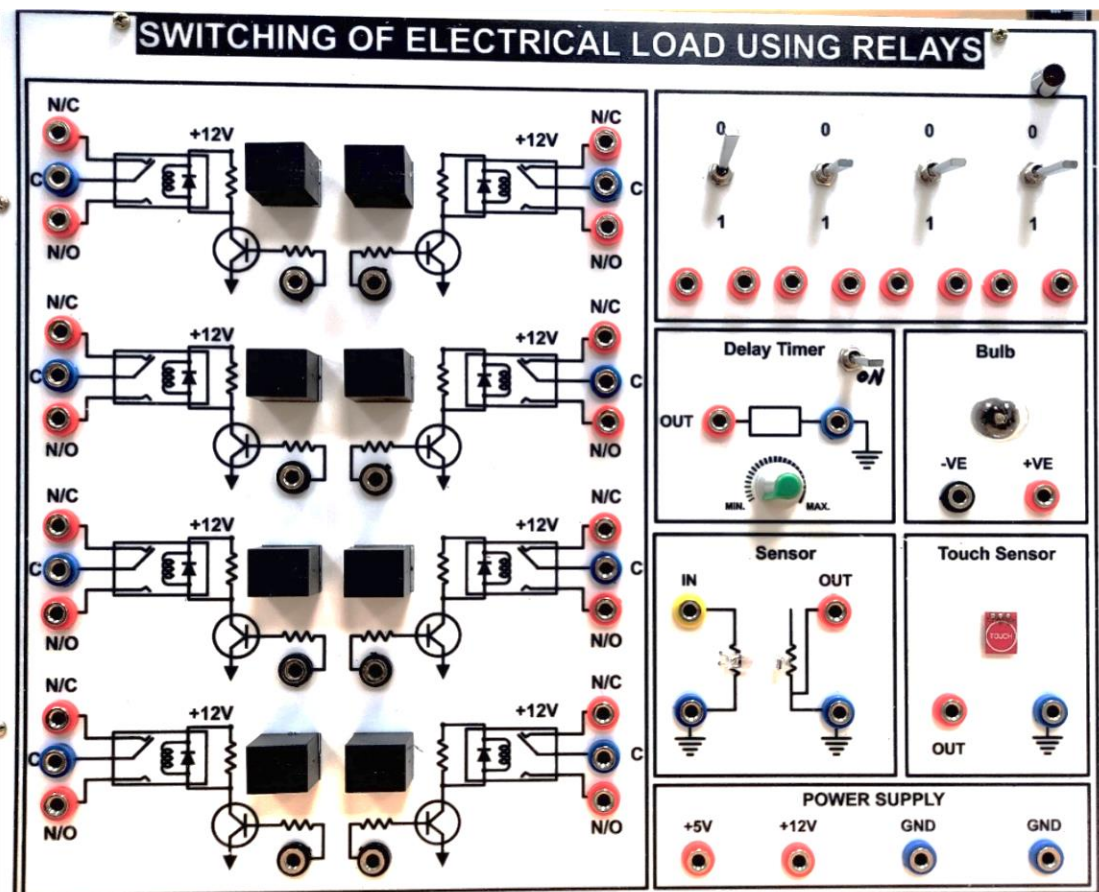


Figure 5 Saturation region

Then we can define the “saturation region” or “ON mode” when using a bipolar transistor as a switch as being, both junctions forward biased, $V_B > 0.7\text{V}$ and $I_C = \text{Maximum}$. The transistor operates as a “single-pole single-throw” (SPST) solid state switch. With a zero signal applied to the Base of the transistor it turns “OFF” acting like an open switch and zero collector current flows. With a positive signal applied to the Base of the transistor it turns “ON” acting like a closed switch and maximum circuit current flows through the device. The simplest way to switch moderate to high amounts of power is to use the transistor with an open-collector output and the transistors Emitter terminal connected directly to ground. When used in this way, the transistors open collector output can thus “sink” an externally supplied voltage to ground thereby controlling any connected load.

Procedure:

1. Use the Trainer as shown in the Figure below, to switch a bulb (load) of +12V, with input from 3 different sensors i.e. Touch sensor, Delay Timer and Optical sensor.



2. Construct the wire diagram of interfacing the touch sensor, delay timer and optical sensor with the relay and transistor.

DELAY TIMER

OPTICAL SENSOR

TOUCH SENSOR



Date: _____

LABORATORY SKILLS ASSESMENT (Psychomotor)

Total Marks: 100

Criteria (Max Marks)	Level 1 0% ≤ S < 50%	Level 2 50% ≤ S < 70%	Level 3 70% ≤ S < 90%	Level 4 90% ≤ S ≤ 100%	Score (S)
Procedural Awareness (20)	Selects inappropriate skills and/or strategies required by the task	Selects and applies appropriate skills and/or strategies required by the task with some errors	Selects and applies the appropriate strategies and/or skills specific to the task without significant errors	Selects and applies appropriate strategies and/or skills specific to the task without any error	
Practical Implementation (30)	Makes several critical errors in applying procedural knowledge of Switching of Electrical Loads Using Relays	Makes few critical errors in applying procedural knowledge of Switching of Electrical Loads Using Relays	Makes some non-critical errors in applying procedural knowledge of Switching of Electrical Loads Using Relays	Applies the procedural knowledge of Switching of Electrical Loads Using Relays in perfect ways	
Safety (10)	Requires constant reminders to follow safety procedures	Requires some reminders to follow safety procedures	Follows safety procedures with only minimal reminders	Routinely follows safety procedures	
Use of Tool/Equipment (20)	Uses tools, equipment and materials with limited competence	Uses tools, equipment and materials with some competence	Uses tools, equipment and materials with considerable competence	Uses tools, equipment and materials with a high degree of competence	
Participation to Achieve Group Goals (10)	Shows little commitment to group goals and fails to perform assigned roles	Demonstrates commitment to group goals, but has difficulty performing assigned roles	Demonstrates commitment to group goals and carries out assigned roles effectively	Actively helps to identify group goals and works effectively to meet them in all roles assumed	
Interpersonal Skills in Group Work (10)	Rarely interacts positively within a group, even with prompting	Interacts with other group members if prompted	Interacts with all group members spontaneously	Interacts positively with all group members and encourages such interaction in others	
Marks Obtained					

Instructor's Signature: _____

Date: _____

LABORATORY SKILLS ASSESSMENT (Affective)

Total Marks: 40

Criteria (Max. Marks)	Level 1 $0\% \leq S < 50\%$	Level 2 $50\% \leq S < 70\%$	Level 3 $70\% \leq S < 90\%$	Level 4 $90\% \leq S \leq 100\%$	Score (S)
Introduction (5)	Very little background information provided or information is incorrect	Introduction is brief with some minor mistakes	Introduction is nearly complete, missing some minor points	Introduction complete and well-written; provides all necessary background principles for the experiment	
Procedure (5)	Many stages of the procedure are not entered on the lab report.	Many stages of the procedure are entered on the lab report.	The procedure could be more efficiently designed but most stages of the procedure are entered on the lab report.	The procedure is well designed and all stages of the procedure are entered on the lab report.	
Data Record (10)	Data is brief and missing significant pieces of information.	Data provides some significant information and has few critical mistakes.	Data is almost complete but has some minor mistakes.	Data is complete and relevant. Tables with units are provided. Graphs are labeled. All questions are answered correctly.	
Data Analysis (10)	Data is presented in very unclear manner. Error analysis is not included.	Data is presented in ways (charts, tables, graphs) that are not clear enough. Error analysis is included.	Data is presented in ways (charts, tables, graphs) that can be understood and interpreted. Error analysis is included.	Data are presented in ways (charts, tables, graphs) that best facilitate understanding and interpretation. Error analysis is included.	
Report Quality (10)	Report contains many errors.	Report is somewhat organized with some spelling or grammatical errors.	Report is well organized and cohesive but contains some grammatical errors.	Report is well organized and cohesive and contains no grammatical errors. Presentation seems polished.	
Marks Obtained					

LABORATORY SKILLS ASSESSMENT (Cognitive)

Total Marks: 10

(If any)	
Marks Obtained	

Instructor's Signature: _____

Date: _____